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Greater effects by performing a small number of eccentric contractions daily than a larger number of them once a week

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Our previous study found that one maximal voluntary eccentric contraction (MVC-ECC) performed daily for 5 days a week for 4 weeks increased MVC-ECC, isometric (MVC-ISO), and concentric contraction (MVC-CON) torque of the elbow flexors more than 10%. The present study investigated the effects of six maximal voluntary eccentric contractions on the MVC torques and biceps brachii and brachialis muscle thickness (MT). Thirty-six healthy young adults were placed to one of the three groups ($N = 12$ per group); the 6×1 group that performed one set of six contractions once a week, the 6×5 group that performed one set of six contractions a day for 5 days a week, and the 30×1 group that performed five sets of six contractions a day in a week. The training duration was 4 weeks for all groups, and changes in MVC-ECC, MVC-CON and MVC-ISO torque, and MT before and after the 4-week training were compared among the groups. The 6×1 group did not show significant changes in muscle strength and MT. Significant ($p < 0.05$) increases in MVC-ECC (13.5 ± 11.5%), MVC-ISO (9.3 ± 5.5%), MVC-CON torque (11.1 ± 7.4%) were evident for the 6×5 group only, and increases in MT were found for the 6×5 (10.4 ± 4.4%) and 30×1 (8.0 ± 5.8%) groups without a significant difference. These results suggest that performing a small number of eccentric contractions 5 days a week is more effective for increasing muscle strength than performing a larger volume of eccentric contractions once a week. However, it appears that training volume is a factor for muscle hypertrophy in a short-term training.

KEYWORDS

elbow flexors, lengthening contraction, maximal voluntary contraction torque, muscle thickness, training frequency

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1 | INTRODUCTION

The effects of resistance training on muscle function and muscle hypertrophy are determined by factors such as muscle contraction type, intensity, number of contractions, and frequency.[1] Regarding the muscle contraction type, previous studies have shown that resistance training focusing on eccentric (lengthening) muscle contractions (actions) is more effective than that focusing on concentric (shortening) muscle contractions (actions) for increasing muscle strength and muscle size.[2,3] We have recently reported that one 3-second maximal eccentric contraction performed daily for 5 days a week (Monday–Friday) for 4 weeks increased maximum voluntary isometric (10.2 ± 6.4%), concentric (12.8 ± 9.6%), and eccentric contraction torque (12.2 ± 7.8%) of the elbow flexors without significant changes in muscle thickness of biceps brachii and brachialis.[4] It is interesting that only a small number of eccentric contractions (20 contractions in 4 weeks) can still increase muscle strength, but it is not known whether a larger number of eccentric contractions per session or in a week produce greater effects on muscle strength and muscle hypertrophy.

To increase the stimulus to the muscle in resistance exercise training, the number of contractions, the number of training sessions per week (frequency), or both can be increased. Since our previous study[4] showed that one 3-second maximal eccentric contraction daily for 5 days a week was effective for increasing muscle strength, it is interesting to examine if a similar number of contractions performed in 1 day a week can still increase muscle strength in a similar magnitude. Grgic et al.[5] showed in their meta-analysis study that increasing the frequency of training could increase muscle strength and muscle size greater. Therefore, it may be that six maximal eccentric contractions daily for 5 days a week increase maximum voluntary isometric, contraction, and eccentric contraction torque of the elbow flexors greater and induce muscle hypertrophy when compared with the protocol of one maximal eccentric contraction a day in the previous study.[4] It is also interesting to investigate if 30 (five sets of six) maximal eccentric contractions performed 1 day a week will induce similar muscle strength gain to the protocol of six maximal eccentric contractions daily for 5 days a week.

Therefore, we compared the effect of three protocols of maximal eccentric contraction training: six maximal eccentric contractions in 1 day per week, six maximal eccentric contractions in 1 day for 5 days per week, and five sets of six maximal eccentric contractions in 1 day per week on changes in maximal voluntary isometric, concentric, and eccentric contraction toque of the elbow flexors and muscle thickness of biceps brachii plus brachialis. We hypothesized that no significant increases in maximal voluntary contraction (MVC) torque and muscle thickness would be found after the training of six maximal eccentric contractions in 1 day per week, but the training of six maximal eccentric contractions in 1 day for 5 days per week, and five sets of six maximal eccentric contractions in 1 day per week (30 eccentric contractions per week) would show increases in both MVC torque and muscle thickness.

2 | MATERIALS AND METHODS

2.1 | Participants and study design

A total of 36 (24 male and 12 female) healthy university students were recruited for this study. All participants were free from orthopedic disorders of the upper extremities, had no history of neuromuscular or chronic diseases, and had not performed any regular resistance training of the arms in the past 6 months before the present study participation. They were placed to one of the three groups (n = 12 per group) with a different number of eccentric contractions in a session and a training frequency in a week; one set of six contractions in a session and 1 day a week (6 × 1 group; the first number represents the number of eccentric contractions in a week, and the second number represents the number of training sessions per week), one set of six contractions in a session and 5 days a week (6 × 5 group), or five sets of six contractions in a session, 1 day a week (30 × 1 group), and the training duration was 4 weeks for all groups (Figure 1). There were no significant differences in the number of males and females in a group, average age, height, and body mass among the groups (6 × 1 group: 8 males, 4 females, 21.5 ± 0.5 y, 165.8 ± 8.1 cm, 58.5 ± 8.9 kg; 6 × 5 group: 7 males, 5 females, 21.1 ± 1.3 y, 165.8 ± 7.8 cm, 62.9 ± 11.8 kg; 30 × 1 group: 9 males, 3 females, 21.1 ± 0.5 y, 171.9 ± 8.3 cm, 64.4 ± 9.6 kg).

The sample size was estimated from our previous study[4] showing 12% increase in maximal voluntary eccentric contraction (MVC-ECC) torque of the elbow flexors after performing a single three-second maximal eccentric contraction a day for 5 days a week for 4 weeks. Since we had not known a possible difference between the groups, we used an effect size based on the above to estimate the number of samples for a group to be enough for detecting a possible increase in MVC-ECC torque after training. Based on the effect size of 0.39 for a possible change in the MVC-ECC torque from pre- and post-intervention, the sample size was estimated to be 12 participants in a group with an alpha level of 0.05 and a power ($1 - \beta$) of 0.8 (G*Power 3.1, Germany). It was assumed that the sample size was adequate to test the hypothesis that no significant

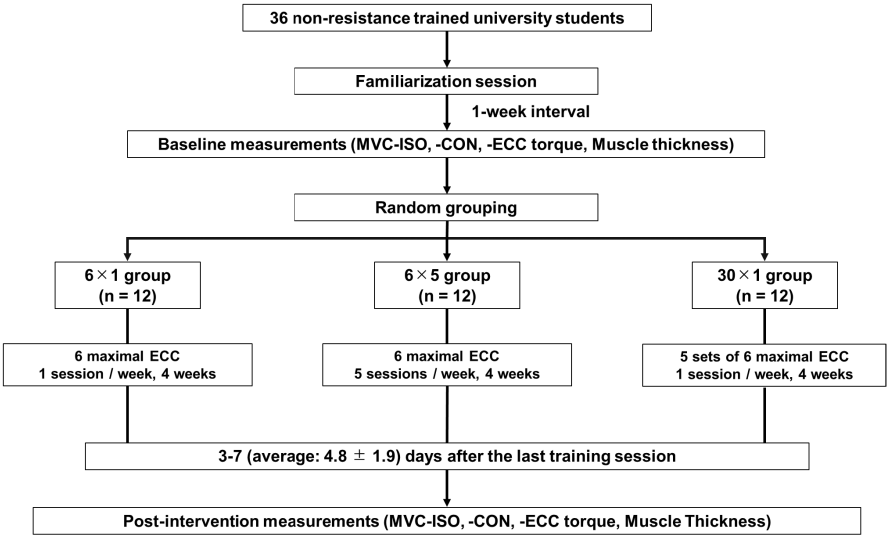


FIGURE 1 Experimental design and protocol. A total of 36 healthy university students were placed to one of the three training groups; 6×1 that performed one set of 6 maximal eccentric contractions in 1 day a week, 6×5 that performed one set of 6 maximal eccentric contractions a day for 5 days a week, and 30×1 that performed 5 sets of 6 maximal eccentric contractions in 1 day a week ($n = 12$ per group), and the intervention duration was 4 weeks for all groups. A familiarization session was held 1 week before the baseline assessments. Before the first intervention session and several days after the last intervention session, maximum voluntary isometric, concentric and eccentric contraction (MVC-ISO, MVC-CON, and MVC-ECC) torque and muscle thickness of the biceps brachii and brachialis was measured

increases in MVC torque and muscle thickness would be found for the 6×1 group, but the 6×5 and 30×1 group would show increases in both MVC torque and muscle thickness.

All participants were informed about the study purpose and procedures, and a written consent was obtained from each participant. This study was approved by the Ethics Committee of Niigata University of Health and Welfare (#18305). The study was conducted in conformity with the policy statement regarding the use of human subjects by the Declaration of Helsinki.

The dependent variables consisted of maximal voluntary isometric contraction (MVC-ISO) torque at three different elbow joint angles (20°, 55°, 90°), concentric (MVC-CON), and eccentric contraction (MVC-ECC) torque at two angular velocities (30°/s, 180°/s), and biceps brachii and brachialis muscle thickness. Several MVC measurements were taken in accordance with our previous study[4] to examine whether any specifity of the training would be observed for other measurements than MVC-ECC torque. These measurements were taken from the trained arm only, thus we did not investigate the cross-education effect in the present study. Changes in these variables from pre- to post-intervention (i.e., the 4-week training) were compared among the three training groups.

A familiarization session was set 1 week before the baseline measurements in which all participants performed the MVC-ISO, MVC-CON, and MVC-ECC torque measurements (Figure 1). The torque measure protocols in the familiarization were the same as those for the baseline measures described below. Thus, the number of maximal voluntary contractions was six for isometric, six for concentric, and six for eccentric torque measures. The baseline measures were taken 1 week after the familiarization session, and the post-training measurements were taken between three and seven (average: 4.8 ± 1.9) days after the last training session (the range was due to the COVID-19 restrictions). All participants were instructed to refrain from any systematic training outside the study during the experiment period.

2.2 | Training protocol

The participants performed six maximal voluntary eccentric contractions of the elbow flexors using their dominant arms on an isokinetic dynamometer (Biodex System 3.0, Biodex Medical Systems Inc., Shirley, NY, USA) either one set a week (6 ×1 group), one set a day for 5 days a week (6 ×5 group), or five sets a day for 1 day a week (30 ×1 group). The arm dominance was determined by the Edinburgh Handedness Inventory.[6] In the eccentric contractions, the shoulder joint was set at 45° flexion, and the trunk and pelvis were fixed with belts to the dynamometer chair. The participants were instructed to resist maximally against the movement of the dynamometer lever arm moving from 100° to 10° of elbow flexion at the angular velocity of 30°/s. After each eccentric contraction, the lever arm of the isokinetic dynamometer passively returned the elbow joint to

the starting position (100° flexion angle) at 10°/s, which provided a 9-s rest between contractions. For the 30 × 1 group, a 120-s interval was inserted between sets. Verbal encouragement was provided to the participants to generate maximal force.

Before each training session, muscle soreness of the elbow flexors was assessed by a visual analog scale (VAS) consisting of a 100-mm line with 0 for no pain and 100 for the maximum pain imaginable, when the participant extended the elbow joint maximally.[6]

2.3 | Dependent variables

2.3.1 | Maximum voluntary contraction (MVC) torque

According to the previous study,[4] the MVC torque was measured at 20° (MVC-ISO_{20}), 55° (MVC-ISO_{55}), and 90° (MVC-ISO_{90}) of elbow flexion in the same setting as that of the training using the isokinetic dynamometer. The MVC-ISO torque measurements for each angle were performed in random order. Each contraction lasted for 3 seconds, two measurements were taken for each angle with a 45-s interval, and the larger value of the two measures was used for further analysis. After the MVC-ISO torque measures, MVC torque of concentric (MVC-CON) and eccentric contraction (MVC-ECC) of the elbow flexors were measured in the same setting of the dynamometer to that of the MVC-ISO torque measures. MVC-CON torque was measured at 30°/s (MVC-CON_{30}) and 180°/s (MVC-CON_{180}), and MVC-ECC torque was also measured at 30°/s (MVC-ECC_{30}) and 180°/s (MVC-ECC_{180}) in this order. The rest time between measurements was 120 s. The range of motion was 100° for the MVC-CON and MVC-ECC torque measurements, the starting angle was 10° for MVC-CON, and 100° elbow flexion for MVC-ECC, where the fully extended elbow joint was defined as 0°. In the case of MVC-CON torque measurement, each participant was instructed to perform MVC from 10° to 100° of elbow flexion only, and the arm was returned to the starting angle (10° elbow flexion) passively while relaxing. For the MVC-ECC torque measurement, after performing eccentric MVC from 100° to 10° of elbow flexion, the arm was returned to the starting angle (100° elbow flexion) in a passive motion. In the MVC-CON torque measure, each participant performed MVC three times consecutively with a 120-s rest between contractions for each angular velocity, and the maximum torque obtained was used for the subsequent analysis. This was also the case for the MVC-ECC measures. During all measurements, verbal encouragement was provided to the participants. The torque of each contraction was monitored and recorded via an analog-to-digital converter (PowerLab 8/30, AD Instruments, Colorado Springs, CO, USA) to a personal computer with the analysis software (LabChart 7, AD Instruments).

As we had done in our previous study,[4] the average of MVC-ISO_{20}, MVC-ISO_{55}, and MVC-ISO_{90} was calculated as MVC-ISO_{ave} and used for further analysis. In addition, the average of MVC-CON_{30} and MVC-CON_{180} torque was calculated as MVC-CON_{ave} torque, and the average of MVC-ECC_{30} and MVC-ECC_{180} torque was also calculated as MVC-ECC_{ave} torque and used for further analysis. Then, the mean value of MVC-ISO_{ave}, MVC-$CON_{ave,}$ and MVC-ECC_{ave} was obtained (MVC_{ave}) for each participant. This made us possible to compare the results of the present study with those of the previous study,[4] which was done in the discussion.

Based on our previous study,[4] the test–retest reliability (coefficient of variation) for the measures separated by 4 weeks without any training was 4.0% for MVC-ISO_{20}, 4.1% for MVC-ISO_{55}, 2.2% for MVC-ISO_{90}, 5.6% for MVC-CON_{30}, 4.5% for MVC-CON_{180}, 4.0% for MVC-ECC_{30}, and 2.8% for MVC-ECC_{180} torque.

2.3.2 | Muscle thickness

A total of biceps brachii and brachialis muscle thickness (MT) of the dominant (trained) arm was measured using B-mode ultrasonography with an 8-MHz linear probe (LOGIQ e V2; GE Healthcare Japan, Tokyo, Japan). The ultrasound intensity was 78.0, the frequency was 8.0 MHz, the depth was 4.0 cm, which were consistent over measurements at different time points across participants. The investigator minimized the probe's pressure against the skin as much as possible, and the same investigator took all measurements. The measurement sites were 50%, 60%, and 70% of the lateral epicondyle of the humerus from the acromion. Each participant lay in the supine position on a bed with the dominant arm placed at each side with the forearm being supinated while relaxing the arm. Ultrasound measurements of the transverse-axis were repeated twice, and the MT of biceps brachii plus brachialis was measured as the distance from the inner edge of the fascia to the humerus. The average of the two measurements was used for further analysis. In addition, the average value (MT_{ave}) of the MT at the 50%, 60%, and 70% sites was calculated and used for further analysis.

Based on our previous study,[4] the test–retest reliability (coefficient of variation) for the measures separated by 4 weeks without any training was 1.4% for MT_{50}, 1.0% for $MT_{60,}$ and 0.7% for MT_{70}. The intra-rater reliability for MT50, MT60, and MT70 was 0.974, 0.974, and 0.967, respectively.

2.4 | Statistical analyses

Statistical analyses were performed using the SPSS version 24.0 (IBM Japan, Inc., Tokyo, Japan). The normality of the data was checked by a Shapiro–Wilk test. Group differences in the baseline measures were assessed using a one-way analysis of variance (ANOVA). The sum of the peak torques produced in the training sessions was compared among the three groups by one-way ANOVA first followed by an independent t-test with Bonferroni correction for two group comparisons. A split-plot ANOVA with two factors (group: 6×1 vs. 6×5 vs. 30×1, time: pre- vs. post-training) was used to compare the groups for the changes in the dependent variables. The magnitude of the change in each variable from pre- to post-training was compared between the groups using multiple comparisons with Bonferroni correction. A Pearson product–moment correlation coefficient was used to determine the relationship between the changes in all MVC torque and MT measures in each group. Effect size (ES) was calculated as a difference in the mean values between pre- and post-training divided by the pooled SD. ES of 0.00–0.19 was considered trivial, 0.20–0.49 was small, 0.50–0.79 was moderate, and ≥0.80 was large.[7] The differences were considered statistically significant at an alpha level of 0.05. Descriptive data are shown as mean ± SD.

3 | RESULTS

3.1 | Training

All participants completed the 4-week training intervention as planned. The average (±SD) torque produced in each eccentric contraction for the 6×1, 6×5, and 30×1 groups over training sessions are shown in Figure 2. In the 6×1 group, the peak torque produced in a session decreased ($p<0.001$) over six contractions for the four sessions. A significant interaction effect was found for the changes in the average torque of six contractions over 4 weeks (F [3, 33] = 12.05, $p<0.001$, $\eta_p^2 = 0.523$). The post-hoc test showed that the torque was greater for Week 3 ($p = 0.013$), Week 4 ($p = 0.003$) when compared with Week 1, but no significant differences were evident for Week 2 – Week 4. In the 6×5 group, the peak torque produced in a session decreased ($p<0.001$) over six contractions for all sessions. The average torque of six contractions increased over 4 weeks (F [3, 33] = 10.45, $p<0.001$, $\eta_p^2 = 0.487$), and the post-hoc test showed greater torque in Week 3 ($p = 0.013$) and Week 4 ($p = 0.005$) than Week 1 as well as in Week 3 than Week 2 ($p = 0.019$). In the 30×1 group, the peak torque produced in a session decreased ($p<0.001$) over 30 contractions for the four sessions. The average torque of 30 contractions increased over 4 weeks (F [3, 33] = 6.00, $p = 0.002$, $\eta_p^2 = 0.353$), and the post-hoc test showed the torque was greater for Week 4 than Week 1 ($p = 0.023$), Week 2 ($p = 0.001$), and Week 3 ($p = 0.039$). When comparing the toque produced in the first contraction of each week sessions, the 6×1 and 30×1 groups showed that the torque in the Weeks 2–4 was larger ($p<0.01$) than that in the Week 1, and the 6×5 group also had a greater ($p<0.001$) torque in the Week 3 and Week 4 than Week 1. Regarding muscle soreness before each session, no or little (VAS: 2.0 ± 5.1 mm) muscle soreness was evident for the 6×1 group, and this was also the case for the 6×5 group (5-day average: 0 ± 0–2.6 ± 6.6 mm). In the 30×1 group, the VAS before each of the four sessions was 0 ± 0 mm (Week 1), 9.6 ± 10.3 mm (Week 2), 3.8 ± 5.3 mm (Week 4), and 0.7 ± 1.7 mm (Week 4), and the Week 2 value was greater ($p<0.05$) than those of other weeks. The sum of the peak torques over the 4-week training for the 6×1, 6×5, and 30×1 groups were 1241.5 ± 286.4 N m, 5907.4 ± 1597.5 N m, and 4519.7 ± 1093.8 N m, respectively. The value was greater ($p<0.001$) for the 6×5 and 30×1 groups than the 6×1 group, but no significant difference was evident between the 6×5 and 30×1 groups ($p = 0.08$).

3.2 | MVC torque

A significant interaction effect was evident for changes in MVC-ISO_{55} (F [2, 33] = 3.94, $p = 0.029$, $\eta_p^2 = 0.193$), MVC-ISO_{90} (F [2, 33] = 5.798, $p = 0.007$, $\eta_p^2 = 0.26$), MVC-CON_{30} (F [2, 33] = 8.838, $p = 0.001$, $\eta_p^2 = 0.349$), MVC-CON_{180} (F [2, 33] = 5.851, $p = 0.007$, $\eta_p^2 = 0.262$), MVC-ECC_{30} (F [2, 33] = 5.975, $p = 0.006$, $\eta_p^2 = 0.266$), and MVC-ECC_{180} (F [2, 33] = 6.982, $p = 0.003$, $\eta_p^2 = 0.297$) torque. The post-hoc tests showed that MVC-ISO_{55} ($p = 0.001$, $d = 0.34$), MVC-ISO_{90} ($p = 0.042$, $d = 0.36$), MVC-CON_{30} ($p = 0.008$, $d = 0.39$), MVC-CON_{180} ($p = 0.006$, $d = 0.58$), MVC-ECC_{30} ($p = 0.032$, $d = 0.45$), and MVC-ECC_{180} ($p = 0.022$, $d = 0.52$) torque increased only for the 6×5 group (Figure 3). When looking at individuals, most of the participants in the 6×5 group showed increases in the MVC torques from pre- to post-intervention, but this was not the case for the other two groups.

Significant differences among the groups were also evident for changes in MVC-ISO_{AVE} (F [2, 33] = 6.083, $p = 0.006$, $\eta_p^2 = 0.269$), MVC-CON_{AVE} (F [2, 33] = 10.922, $p<0.01$, $\eta_p^2 = 0.398$), MVC-ECC_{AVE} (F [2, 33] = 7.901, $p = 0.002$, $\eta_p^2 = 0.324$) and MVC_{AVE} torque (F [2, 33] = 14.976, $p<0.01$, $\eta_p^2 = 0.476$) with significant increases only for the 6×5 group revelated by the post-hoc tests. Figure 4 shows normalized changes in the average of MVC-ISO torque at three angles (MVC-ISO_{AVE}), the average of MVC-CON (MVC-CON_{AVE}), and MVC-ECC torque at two angular velocities (MVC-ECC_{AVE}), and the

average of all torque measures (MVC_{AVE}) from pre- to post-intervention for each participant and the average (±SD) of each group. The 6×5 group showed significant increases in MVC-ISO_{AVE} ($p = 0.002$, $d = 0.33$), MVC-CON_{AVE} ($p = 0.001$, $d = 0.48$), MVC-ECC_{AVE} ($p = 0.016$, $d = 0.49$), and MVC_{AVE} torque ($p < 0.001$, $d = 0.44$), but no significant increases in the MVC variables were evident for the 6×1 and 30×1 groups.

3.3 | Muscle thickness (MT)

Figure 5 shows changes in MT at three different sites from pre- to post-intervention for individual participants, and normalized changes in the average MT of the three sites for the three groups. The baseline MT values at 50%, 60%, and 70% sites and their average value (MT_{ave}) were not significantly different among groups. A significant interaction effect was evident for changes in MT at 50% (MT_{50}: F [2, 33] = 18.70, $p < 0.001$, $\eta_p^2 = 0.53$), 60% (MT_{60}: F [2, 33] = 19.35, $p < 0.001$, $\eta_p^2 = 0.54$), 70% (MT_{70}: F [2, 33] = 18.72, $p < 0.001$, $\eta_p^2 = 0.53$), and MT_{ave} (F [2, 33] = 25.73, $p < 0.001$, $\eta_p^2 = 0.60$). The post hoc test showed that the 6×5 and 30×1 groups showed increases at 50% ($p = 0.001$, $d = 0.53$; $p = 0.003$, $d = 0.46$), 60% ($p < 0.001$, $d = 0.49$; $p = 0.001$, $d = 0.58$), and 70% ($p < 0.001$, $d = 0.46$; $p = 0.014$, $d = 0.54$) sites. The MT_{ave} also showed that the 6×5 and 30×1 groups had a significant increase by

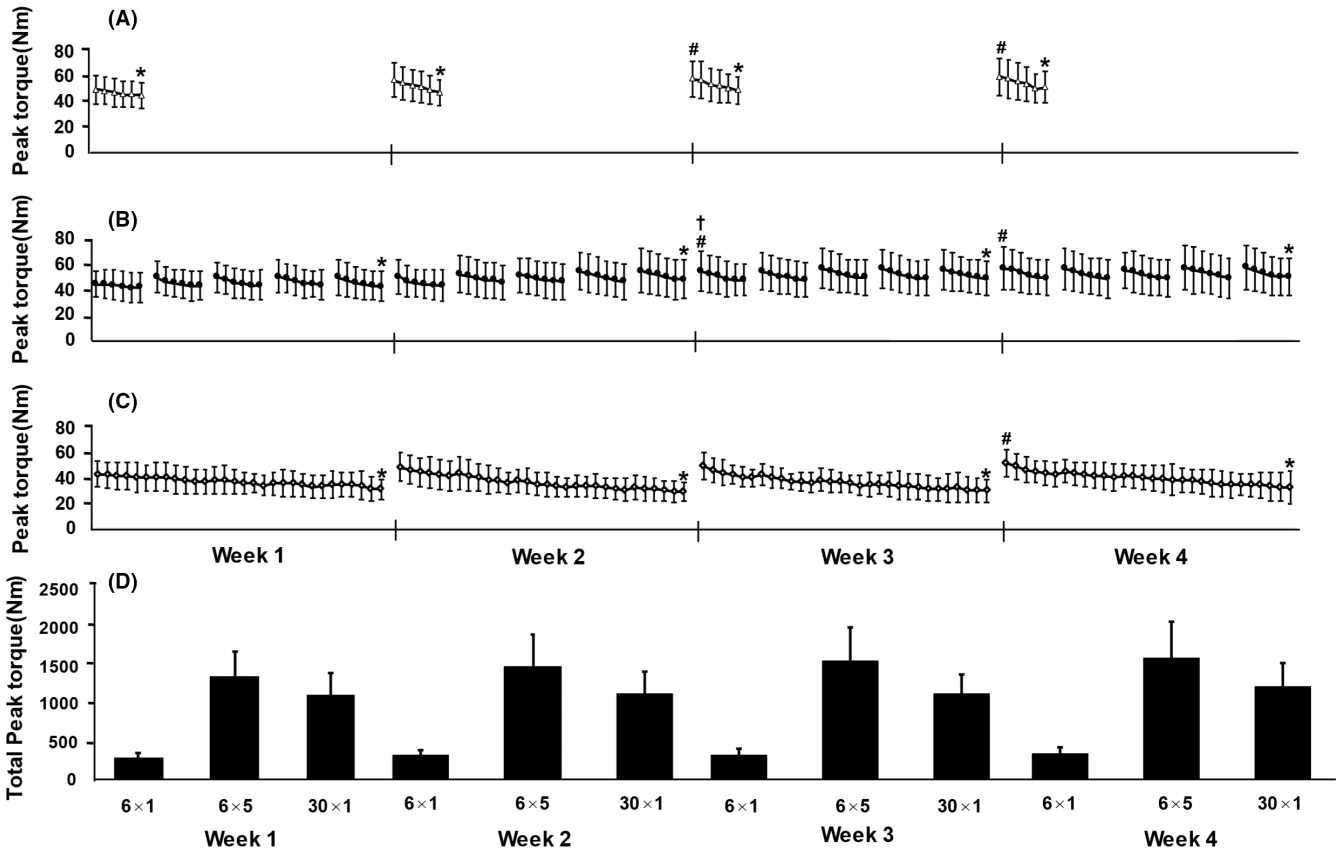
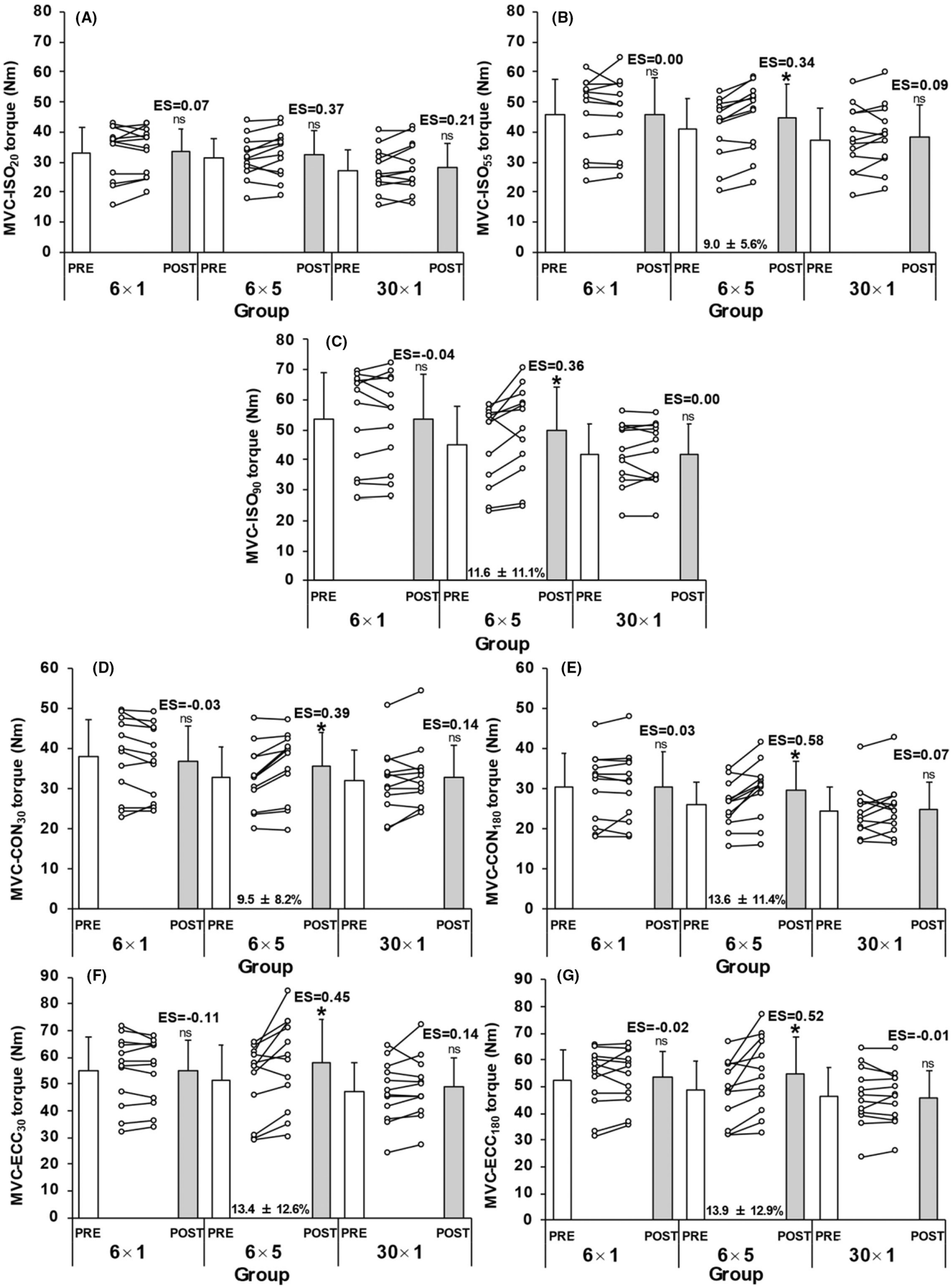


FIGURE 2 Changes (mean ± SD of 12 participants) in maximum voluntary eccentric contraction torque in one set of 6 maximal eccentric contractions in 1 day a week (6×1 group: A), one set of 6 maximal eccentric contractions a day for 5 days a week (6×5 group: B), and 5 sets of 6 maximal eccentric contractions in 1 day a week (30×1 group: C), and total peak torque for each week (D) over training sessions intervention period. *: Significantly ($p < 0.05$) decreased from first contraction torque in the set, #: significantly ($p < 0.05$) different from the Week 1 at first contraction torque, †: significantly ($p < 0.05$) different from the Week 2 at first contraction torque.

FIGURE 3 Changes in maximum voluntary isometric contraction torque at three different angles (20°: A, 55°: B, and 90°: C), maximum voluntary isokinetic torque at two velocities (30°/s, 180°/s) for concentric (D, E) and eccentric contraction of the elbow flexors (F, G) from baseline (PRE) to post-intervention (POST) for the individuals and their average (±SD) of the group that performed one set of 6 maximal eccentric contractions in 1 day a week (6×1 group), one set of 6 maximal eccentric contractions a day for 5 days a week (6×5 group), and 5 sets of 6 maximal eccentric contractions in 1 day a week (30×1 group) for 4 weeks. *: significantly ($p < 0.05$) different from the PRE value. ns: no significant difference from the PRE value. The average (±SD) magnitude of change (%) is included for the significant ($p < 0.05$) change and Cohen's d effect size (ES) is also provided for all groups.


(A)
MVC-ISO20 torque (Nm)
ES=0.07
ns
ES=0.37
ns
ES=0.21
ns
PRE
POST
6×1
6×5
30×1
Group
(B)
MVC-ISO55 torque (Nm)
ES=0.00
ns
ES=0.34
*
ES=0.09
ns
9.0 ± 5.6%
(C)
MVC-ISO90 torque (Nm)
ES=-0.04
ns
ES=0.36
*
ES=0.00
ns
11.6 ± 11.1%
(D)
MVC-CON30 torque (Nm)
ES=-0.03
ns
ES=0.39
*
ES=0.14
ns
9.5 ± 8.2%
(E)
MVC-CON180 torque (Nm)
ES=0.03
ns
ES=0.58
*
ES=0.07
ns
13.6 ± 11.4%
(F)
MVC-ECC30 torque (Nm)
ES=-0.11
ns
ES=0.45
*
ES=0.14
ns
13.4 ± 12.6%
(G)
MVC-ECC180 torque (Nm)
ES=-0.02
ns
ES=0.52
*
ES=-0.01
ns
13.9 ± 12.9%

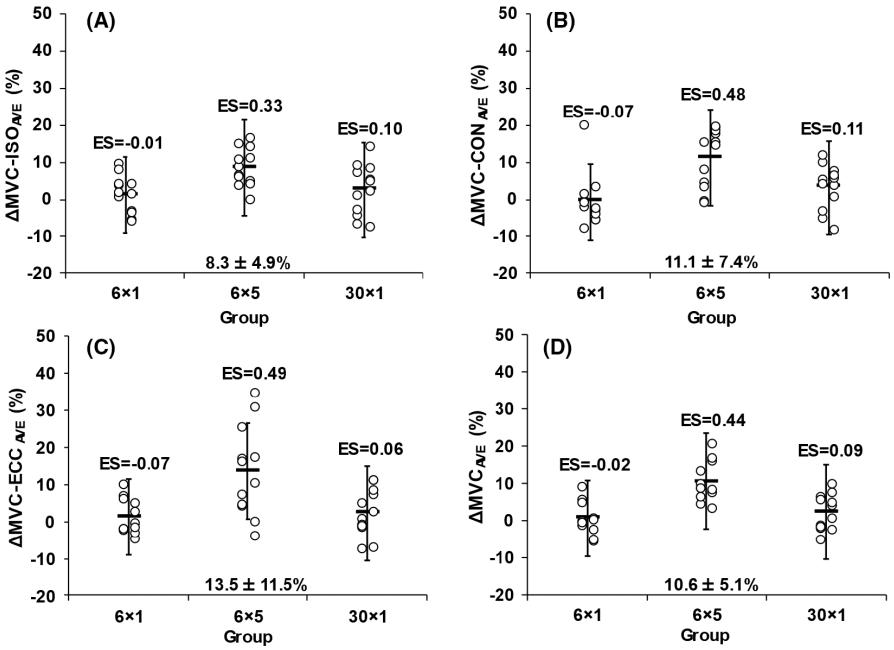


FIGURE 4 Normalized changes in average maximum voluntary isometric contraction torque at three different angles (MVC-ISO$_{AVE}$: A), and average maximum voluntary concentric (MVC-CON$_{AVE}$: B) and eccentric contraction torque at two different velocities (MVC-ECC$_{AVE}$: C), and the average of all torque measures (MVC$_{AVE}$: D) from baseline to post-intervention for the individuals and their average (±SD) of the group that performed one set of 6 maximal eccentric contractions in 1 day a week (6×1 group), one set of 6 maximal eccentric contractions a day for 5 days a week (6×5 group), and 5 sets of 6 maximal eccentric contractions in 1 day a week (30×1 group) for 4 weeks. *: significantly ($p < 0.05$) different from the PRE value. ns: no significant difference from the PRE value. The average (±SD) magnitude of change (%) is included for the significant ($p < 0.05$) change and Cohen's d effect size (ES) is also provided for all groups.

10.3 ± 4.4% ($p < 0.001$, $d = 0.50$) and 8.0 ± 5.8% ($p = 0.002$, $d = 0.54$), respectively, without a significant difference ($p = 0.40$) between the groups, but no significant increase was evident for the 6×1 group.

4 | DISCUSSION

The present study tested the hypothesis that the 6×5 and 30×1 groups but not the 6×1 group would show increases in the MVC torque and MT. The results showed no significant increases in MVC-ISO, MVC-CON, and MVC-ECC torque and MT for the 6×1 group, which was in line with the hypothesis. The 6×5 group showed significant increases in all torque measures and MT, but the 30×1 group had an increase in MT without increases in the MVC torques (Figures 3–5). These results did not support the hypothesis that the 30×1 group would show increases in the MVC torque.

4.1 | MVC torque

Hettinger and Müller[8] reported that once-daily maximal voluntary isometric contraction for 6 seconds increased maximal voluntary isometric contraction strength of the elbow flexors by 5% per week. Walters et al.[9] showed that once-daily maximal voluntary isometric contraction for 15 seconds at 2/3 of the maximal intensity performed over 8 days (5 consecutive days, 2 days off, 3 consecutive days) increased maximal voluntary isometric contraction strength of the elbow flexors by 22%. In contrast, our previous study[4] did not find significant increases in maximal voluntary isometric contraction torque after performing one 3-second maximal isometric or concentric contraction of the elbow flexors for 5 days a week for 4 weeks. However, the one 3-second maximal eccentric contraction of the elbow flexors performed daily for 5 days a week for 4 weeks, increased MVC-ISO (10.2 ± 6.4%), MVC-CON (12.8 ± 9.6%), and MVC-ECC torque of the elbow flexors (12.2 ± 7.8%) without a change in muscle thickness. This was why the present study focused on eccentric contraction training, and the main research questions were (1) whether six maximal eccentric contractions in 1 day per week over 4 weeks would increase MVC-ISO, MVC-CON, and MVC-ECC torque of the elbow flexors; (2) whether six maximal eccentric contractions in 1 day for 5 days per week and five sets of six maximal eccentric contractions in 1 day per week would similarly increase MVC-ISO, MVC-CON, and MVC-ECC torque of the elbow flexors and muscle thickness of biceps brachii plus brachialis.

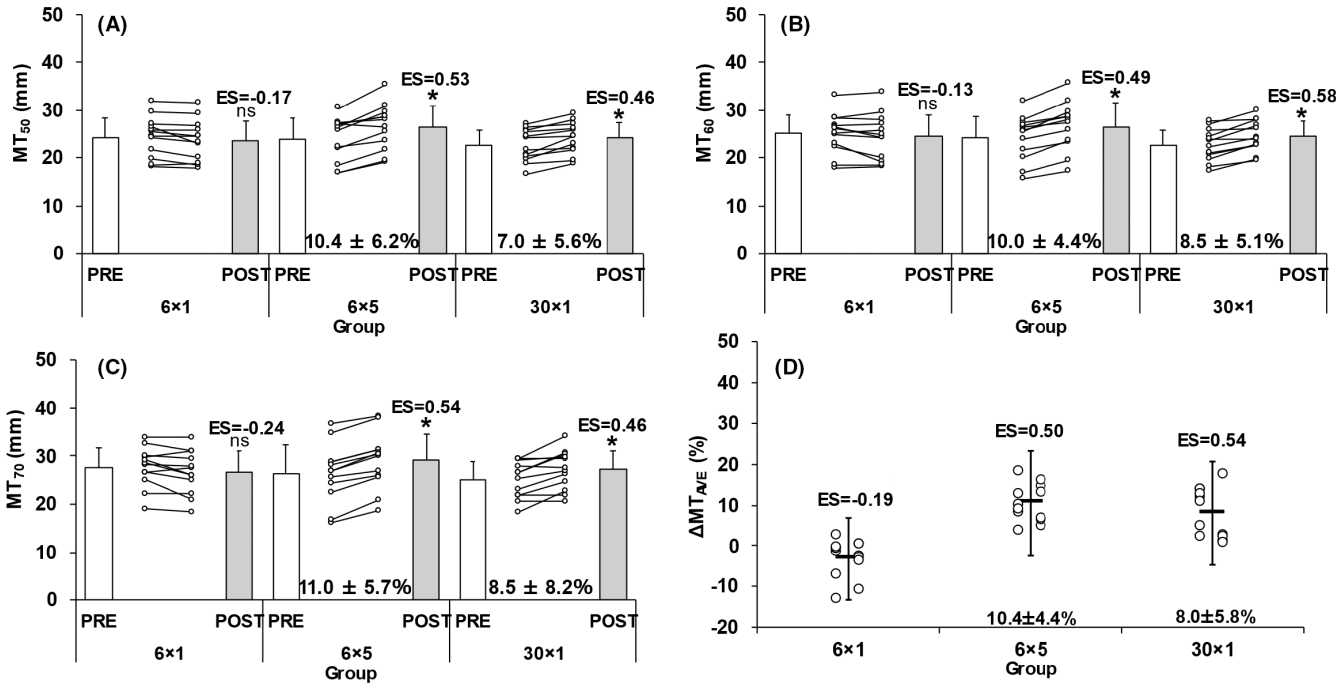


FIGURE 5 Changes in biceps brachii plus brachialis muscle thickness (MT) at three different sites between the lateral epicondyle of the humerus from the acromion (50%: A, 60%: B, and 70%: C), and normalized changes in the average MT of the three sites from baseline to post-intervention for the individuals and their average (±SD) in the group that performed one set of 6 maximal eccentric contractions in 1 day a week (6×1 group), one set of 6 maximal eccentric contractions a day for 5 days a week (6×5 group), and 5 sets of 6 maximal eccentric contractions in 1 day a week (30×1 group) for 4 weeks. *: significantly ($p < 0.05$) different from the PRE value. ns: no significant difference from the PRE value. The average (±SD) magnitude of change (%) is included for the significant ($p < 0.05$) change, and Cohen's d effect size (ES) is also provided for all groups.

The number of maximal eccentric contractions in the previous study[4] was 20 over 4 weeks (one contraction ×5 days ×4 weeks). In the present study, the number of maximal eccentric contractions was 24 over 4 weeks for the 6×1 group. Thus, the total number of eccentric contractions was slightly greater in the present than the previous study.[4] However, no significant increases in MVC-ISO, MVC-CON, and MVC-ECC torque were observed in the 6×1 group (Figure 3). Interestingly, no significant increases in the MVC torques were evident for the 30×1 group either, although the number of maximal eccentric contractions (30 contractions ×4 weeks = 120) was five times larger than that of the 6×1 group and six times larger than that of the previous study.[4] These suggest that the eccentric training does not increase MVC strength if it is performed once a week, even if a larger number of maximal eccentric contractions are performed in a session. On the other hand, significant increases in the MVC torques were evident for the 6×5 group, although the total number of eccentric contractions performed over 4 weeks was the same as that of the 30×1 group. It appears that to increase MVC torques by eccentric resistance training, it is better to perform a small number of maximal eccentric contractions daily than a larger number of maximal eccentric contractions only once a week. Mathews et al.[10] investigated the frequency effect of three maximal voluntary isometric contraction training of the elbow flexors by comparing two, three, four, and five sessions a week over 4 weeks. They reported that the five times a week protocol was most effective for the strength gains. It appears that this is also the case for the maximal voluntary eccentric contraction training.

As shown in Figure 2, peak torque over 30 eccentric contractions in the 30×1 group was significantly decreased in the four sessions, and the magnitude of decrease from the first contraction to the 30th contraction was 27.3% (Week 1), 38.8% (Week 2), 38.5% (Week 3), and 36.6% (Week 4). In contrast, the peak torque decreased smaller over six contractions (4 weeks averge:15.5%), and when compared with the first contraction in the first set and the last contraction of the fifth set, the magnitude of decrease was between 3.6% (Week 1) and 11.5% (Week 4). Although no significant difference in the total peak torque was evident between the 6×5 and 30×1 groups (Figure 2), it appears that fatigue was smaller for the 6×5 than the 30×1 group.

In recent years, a cluster set training has been shown to be more effective than a traditional resistance training to increase muscle strength.[11] It is possible to consider that the 6×5 group had a cluster of six maximal eccentric

contractions with an ample rest for five sets over 5 days. In contrast, the 30×1 group had continuous 30 maximal eccentric contractions with fatigue. When the time between sessions is too long (i.e., 1 week), the muscle adaptations produced after one training session may be lost before the next training session; thus, the training effects may not be accumulated. This may be the case for the 6×1 and 30×1 groups.

As shown in Figure 3, the 6×5 group had increases in all torques except for MVC-ISO$_{20}$, and the average of MVC-ISO (9.3±5.5%), MVC-CON (11.1±7.4%), and MVC-ECC torque (13.5±11.5%) all showed a significant increase. When accounting for a possible noise of the torque measures (2.2–5.6%), more than 5% increase was considered to be an actual increase. It should be noted that the number of participants who showed more than 5% increases in MVC torque out of 12 participants in the 6×5 group was nine for the MVC-ISO, eight for the MVC-CON and MVC-ECC, and 10 for the average of all MVC measures (Figure 4). Importantly, the number of participants who showed more than 5% increase in any of the torque measures was small for the participants in the 6×1 and 30×1 groups (less than six out of 12 participants).

In the present study and our previous study,[4] multiple torque measures were included to investigate whether changes in MVC torque were different among different contraction modes and different elbow joint angles for the MVC-ISO torque (20°, 55°, 90°). We also included two angular velocities (30°/s, 180°/s) for the MVC-CON and MVC-ECC torque measures, since the movement velocity was slow in the training (30°/s), thus we wanted to examine the effect of the eccentric training on faster velocity contractions. The principle of specificity in training is well known, but it is interesting to note that eccentric resistance training appears to provide more versatile effects on muscle strength gains.[2,4,12] In fact, the present study showed that the resistance training with eccentric-only contractions increased MVC torque of not only eccentric contraction but also concentric contraction at slower and faster velocities and isometric contractions at two different elbow joint angles (Figure 3). It should be noted that MVC-ISO$_{20}$ was the only measure that the 6×5 group did not show a significant increase ($p = 0.09$). However, the effect size for the increase in MV-ISO$_{20}$ (d0.37) was not largely different from those of other measures, and majority of the participants in the 6×5 group showed some increases (Figure 3A). Thus, the non-significant result might be due to less homogenous responses among individuals for the measure than other measures. Overall, it appears that eccentric-only resistance training is effective for improving overall muscle strength, which should be considered for training of individuals including athletes and older adults. This should be investigated further in the future studies.

When comparing the magnitude of increases in the MVC torques in the present study with that in our previous study[4] in which one 3-second MVC-ECC was performed once a day for 5 days a week for 4 weeks by similar participants to those of the present study, no significant difference in the increase in MVC-ISO ($p = 0.307$), MVC-CON ($p = 0.547$), and MVC-ECC toque ($p = 0.689$) was evident. This indicates that the number of maximal eccentric contractions has little effect on the increase in muscle strength when performed daily (i.e., 5 days a week). It should be noted that peak torque produced over six maximal eccentric contractions in each day decreased from the first to the sixth contraction in the four sessions (average: 12.7%) as shown in Figure 2. It seems that increasing the frequency of training per week is important for increasing muscle strength in resistance training using maximal eccentric contractions.[10,13] This may indicate that quality rather than quantity is important to increase muscle strength.

Since the present study did not include neurophysiological measures, it is difficult to discuss the mechanisms underpinning the increases in MVC torque by the maximal eccentric contractions. However, it is likely that the MVC torque increases were due to neural adaptations. It has been well documented that the increases in muscle strength in the early phase of resistance training is mainly due to neural adaptations such as increases in motor unit recruitment, discharge rate, and firing frequency caused by increased excitability of corticospinal tracts and increased number of excitatory synapses in spinal motoneurons.[14] Douglas et al.[15] reported that eccentric resistance training induced greater neural and morphological adaptations when compared with isometric or concentric resistance training. Further studies are required to investigate the mechanisms underpinning the effects of a small number of maximal eccentric contractions performed daily on neural adaptations.

4.2 | Muscle thickness (MT)

No significant increase in biceps brachii plus brachialis muscle thickness was found for the 6×1 group, but the 6×5 group (10.4%) and 30×1 group (8.0%) groups showed significant increases in the muscle thickness without a significant difference between the 6×5 and 30×1 groups (Figure 5). When observing individual participants, 10 out of 12 participants in the 6×5 group and seven out of 12 participants in the 30×1 groups showed more than 5% increase in MT (Figure 5). Since a noise of the MT measures was considered to be <1.5%, it is likely

that an increase in MT represents muscle hypertrophy.[16] However, it is also possible that the increase in MT was due to swelling induced by muscle damage.[17] Nosaka et al.[18] reported swelling of biceps brachii and brachialis for several weeks after a bout of 24 maximal eccentric contractions of the elbow flexors. Thus, it might be that the increase in MT for the 30×1 group was due to swelling. However, it appears that muscle damage was minimal even after the first bout of 5 sets of 6 maximal eccentric contractions (Week 1), and no muscle damage was induced in the subsequent three sessions, due to the repeated bout effect.[19] Therefore, it seems that the swelling was not the main reason for the increases in MT. It could have been possible to use echo-intensity to check inflammatory responses,[20] but we were not able to include echo-intensity measures in the present study. Thus, it is not known whether the increase in muscle thickness reflects the accretion and addition of contractile proteins, i.e., muscle hypertropy. To investigate this, an invasive method (i.e., muscle biopsy) is required. It is interesting to examine changes in protein synthesis after an acute and repeated eccentric exercise bouts of the present study protocols.

It is important to note that our previous study, in which one maximal eccentric contraction daily for 5 days a week for 4 weeks, did not observe any increase in the muscle thickness.[4] This suggests that the total number of eccentric contractions should exceed a certain number to increase muscle thickness. Interestingly, the extent of increase in the muscle thickness was similar between the 6×5 and 30×1 groups. This suggests that the total number of eccentric contractions than the training frequency affects muscle thickness increase more. Saric et al.[21] compared two different training frequencies; three versus six times a week, with the same training volume in bench press and squat resistance exercises, and reported that the magnitude of increase in muscle thickness of triceps brachii (more than 10%) and rectus femoris (more than 5%) and vastus medialis (more than 10%) muscles after 6 weeks of the training was similar between the different frequency programs. Schoenfeld et al.[22] reported in their systematic review and meta-analysis article that weekly resistance training frequency did not affect muscle hypertrophy significantly when training volume was equated. These may explain the similar increases in MT between the 6×5 and 30×1 groups.

For muscle hypertrophy to be induced, muscle protein synthesis should be increased and should exceed muscle protein breakdown.[23] If the increases in MT shown by the 6×5 and 30×1 indicate muscle hypertrophy, it seems that muscle protein synthesis increased in the participants of the 6×5 and 30×1 groups. Shad et al.[24] compared daily myofibrillar protein synthesis rates over a 7-day period between low frequency (10 sets of 10 repetitions at 70% one-repetition maximum, once per week) versus high frequency (2 sets of 10 repetitions at ~70% one-repetition maximum, five times per week) resistance exercise training of vastus lateralis. They showed that the myofibrillar protein synthesis rates did not differ between the low frequency (1.46 ± 0.26%/day) and high frequency (1.48 ± 0.33%/day) conditions. Goldberg et al.[25] reported that satellite cells were involved in skeletal muscle hypertrophy. Hyldahl et al.[26] compared a single bout of eccentric versus concentric exercise of the knee extensors for satellite cell activity, and reported that satellite cell content increased 27% after eccentric but not after concentric exercise. Therefore, it is possible that satellite cells were activated similarly between the 6×5 and 30×1 groups. Further studies are needed to investigate the underpinning mechanisms of the MT increases after a relatively small volume of eccentric contractions.

Interestingly, the 30×1 group showed an increase in muscle thickness without an increase in muscle strength. The correlation coefficient between the magnitude of increase in the average of all MVC torques and the magnitude of increase in MT was not significant ($r = -0.12$, $p = 0.69$) for the participants in the 6×5 group. This suggests that the mechanisms underpinning the increases in muscle strength and those for muscle thickness are different. Loenneke et al.[27] stated in their review article that no available evidence supported the claim that changes in muscle size lead to changes in voluntary strength. Several studies[5,28] showed that greater muscle hypertrophy effect was induced by increasing the training volume; however, the effect of muscle hypertrophy could be similar if the training volume per week was the same.[21,29,30] Therefore, it is possible that the same number of maximal eccentric contractions performed over 4 weeks in the 6×5 and 30×1 groups resulted in a similar increase in MT, but increases in muscle strength were evident in the 6×5 group only due to the neural adaptations discussed above. Further studies are warranted to investigate the mechanisms of muscle hypertrophy induced in a relatively short period (i.e., 4 weeks) by eccentric contractions and why the increases in MT were similar between the 6×5 and 30×1 groups.

4.3 | Limitations and future research directions

There are several limitations in the present study, which should be considered when interpreting the findings of the present study and designing future studies. First, the present study used young sedentary adults as participants, and the sample size was not necessarily large, although it

was considered to be adequate. It is necessary to increase the sample size and investigate if similar results are obtained for older adults, children, and clinical populations. Second, the training frequency was set to once or five times per week in the present study. It is interesting to investigate the effects of different frequencies, such as twice or three times a week or 7 days a week, and clarify the appropriate training frequency for muscle strength increase. It is also important to extend the training period (e.g., 12 weeks). Third, the target muscle group was the elbow flexor muscle group in this study. Therefore, it is necessary to examine whether similar results can be obtained for lower limb and trunk muscles. Fourth, the present study showed that 30 maximal eccentric contractions per week over 5 days (six contractions daily for 5 days) or 1 day (five sets of 6 contractions once a week) induced muscle hypertrophy after 4 weeks. It is interesting to examine the minimum number of contractions that could induce muscle hypertrophy. Fifth, the intervention used maximal voluntary eccentric contractions; however, they are difficult to do without an isokinetic dynamometer or other specific devices. Maximal eccentric contractions may not be possible for the elderly or clinical populations, and they may not be safe as well. Therefore, it is necessary to examine the effect of submaximal eccentric contractions such as two-thirds or one-thirds of maximal intensity. Sixth, the participants in the present study were a mixture of males and females, and the number of male participants was greater. Although the comparisons between sexes were not possible due to the small sample size, it did not appear that their responses to the training were vastly different. It is necessary to properly investigate sex differences for the responses to eccentric training. Seventh, the time point for the post-training measurements was originally set at 3 or 4 days after the last training session, but due to the COVID-19 restrictions, the measurements were taken at 5 ($n = 4$), 6 ($n = 6$), 7 ($n = 4$), 8 ($n = 1$), or 9 ($n = 2$) days post-training for some participants. However, these participants were distributed to the three groups almost evenly, and careful examination of these individuals in comparison to those who had the post-training measurements at 3 or 4 days after the last training did not find large the effects of the post-training measurement timing on the outcome measures. Lastly, the neurophysiological and molecular mechanisms underpinning the effects of the maximal eccentric contraction training were not explored in the present study. Thus, mechanistic approach is warranted in future studies.

4.4 | Conclusions

Together with our previous study showing that one maximal voluntary eccentric contraction performed daily for 5 days a week for 4 weeks increased MVC torque of the elbow flexors by more than 10%,[4] the present study showed performing a small number of maximal eccentric contractions (six contractions) on 5 days a week was more effective for increasing muscle strength than performing a larger volume of eccentric contractions once a week (i.e., 30 contractions). However, muscle thickness of the elbow flexors increased similarly (8–10%) for the group that performed six maximal eccentric contractions a day for 5 days and that performed 30 maximal eccentric contractions a day once a week for 4 weeks. It appears that training volume is more important than the frequency for muscle hypertrophy in resistance training.

4.5 | Perspectives

The recommendation for exercise by the World Health Organization[27] and other organizations such as the American College of Sports Medicine[30] is to do at least 150 minutes of moderate-intensity activities in a week and muscle-strengthening activities that involve all major muscle groups on two or more days a week. However, many people do not meet this recommendation. The present study, together with our previous study[4] showed that only one 3-second maximal eccentric contraction a day was effective for increasing muscle strength, and six 3-second maximal eccentric contractions a day were effective for increasing muscle strength and muscle hypertrophy if performed 5 days a week. These suggest that performing a small amount of eccentric resistance exercise more frequently is a key for increasing muscle strength, or at least such protocol is beneficial as a starting point. We have not investigated other muscles yet, but if we find the three-second rule also applies to other muscles, it is possible to do whole-body resistance exercises in a short period of time. One limitation is that we need a special device to do maximal eccentric contractions. It may be that submaximal eccentric contractions can still work, which warrant further studies.

CONFLICT OF INTEREST

All authors declare that they have no conflict of interest.

DATA AVAILABILITY STATEMENT

All data supporting the conclusions of this study will be fully provided on request by the authors.

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